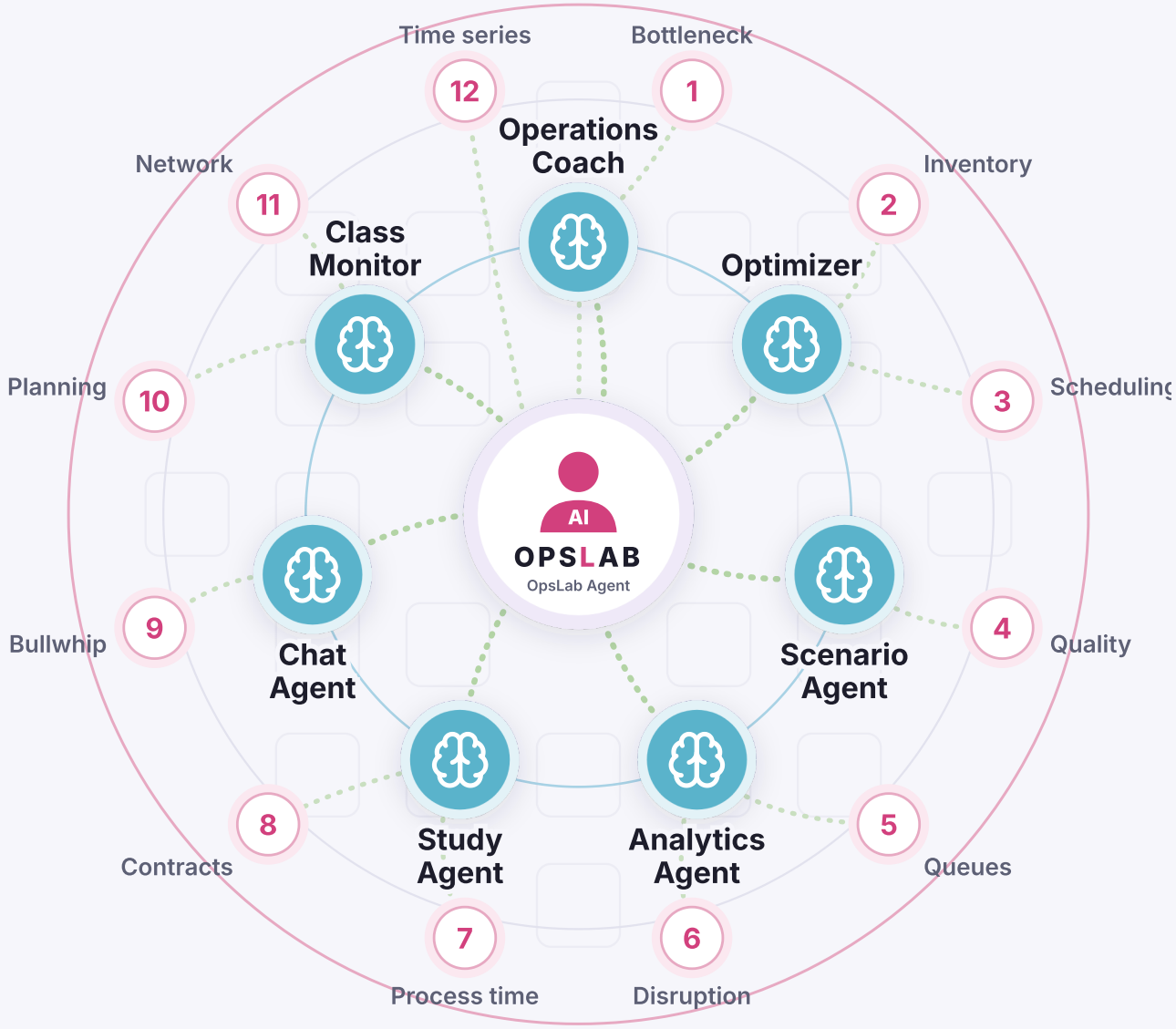
AI-AGENT PRODUCT · OPERATIONS MANAGEMENT

OpsLab: an AI agent team for operations decisions

Students make real operations decisions. A team of AI agents watches, tests better options, explains the trade-offs and coaches them, all inside one web page.



Inner ring: 7 AI agents around the OpsLab Agent hub · **Outer ring:** 12 simulation skills on one shared engine

7	12	100%	0
AI agents working together	simulation skills, from bottlenecks to SARIMA	of numbers computed, never guessed	installs, logins or fees to start

WHY IT IS USEFUL

Learn operations by deciding, not just by reading formulas

Most operations courses teach EOQ, queues or ARIMA as formulas on a slide. Students rarely get to make a decision, see what happens, and learn what would have been better. OpsLab gives every student that loop, with an AI agent team as their coach.

For students

- Play the operations manager in 12 cases
- See results instantly: KPIs, charts, before vs after
- Get a coach that says what went wrong and why
- Turn your session slides into notes and quizzes

For professors

- Run a class game in a 75-minute session
- Students join with a room code and roll number
- Live leaderboard and the class's common mistakes
- Fresh cases and quarterly or yearly campaigns

For programmes

- One HTML file: open it in any browser
- Works offline and costs nothing to run
- Plug in any AI model, including free local Ollama
- Results match Python (statsmodels) for analytics

One session, start to finish



What changes for the learner. Instead of “the answer is EOQ = 219”, students find out that their reorder point of 80 caused 16 stockout days. The agent then shows a policy that cuts total cost from $5,155 to $2,060 and explains why.

What changes for the class. Every run is re-computed by the same engine and scored against the best option the agents found, so discussions move from “who is right?” to “why is this better?”.

MEET THE AGENT TEAM

Seven agents, one goal: better operations decisions

The **OpsLab Agent** sits at the centre and coordinates six specialists. They share one simulation engine, so they all work from the same computed facts.

Operations Coach

Watches every run in silence, then reports: your score, what changed, and a diagnosis of what went wrong, ranked by severity.

After every run · Diagnosis · Session report

Optimizer

Tests hundreds of alternatives on the same scenario (grid, exhaustive and textbook rules) and ranks them, so the best option is proven, not assumed.

After every run · Best option · Gains & give-ups

Scenario Agent

Writes new cases and whole campaigns from a one-line brief, using any connected AI model, and keeps every value inside allowed limits.

On request · New case · Campaign rounds

Analytics Agent

Coaches the time-series lab step by step: stationarity, differencing, ACF/PACF, and fitting ARIMA, SARIMA and smoothing models. It waits for your decision at every step.

Step by step · ADF, ACF, PACF · Model tournament

Study Agent

Reads your session slides and readings, then writes notes, draws concept diagrams, sets quizzes and answers questions with slide references.

On upload · Notes · Quizzes · Help

Chat Agent

Answers free-form questions by calling simulator tools: it runs what-ifs and the optimizer before it replies, then offers an Apply button.

When you ask · What-ifs · Suggested settings

Class Monitor

Runs the class room: re-computes every student run, keeps a fair leaderboard and surfaces the class's most common mistakes for discussion.

During class · Leaderboard · Common mistakes

OpsLab Agent (the hub)

Coordinates the team: it decides which specialist acts, keeps your decision history, and turns their findings into one clear report and a printable session report.

Always on · Works with or without an AI model

HOW THE AGENTS WORK TOGETHER

From your decision to a better one, in six steps

The agents never interrupt. They watch quietly and report after each run. Detailed advice opens only when you ask for it.

1 You decide

Set the decisions in blue, e.g. a reorder point and order size. The scenario (demand, costs, lead time) is the world you face.

2 Engine simulates

The simulation engine plays out 90 days, 52 weeks or 1,000 seasons with fixed random streams, so results are repeatable.

3 Coach diagnoses

The Operations Coach spots what went wrong ("reorder point below lead-time demand") and logs your decision history.

4 Optimizer explores

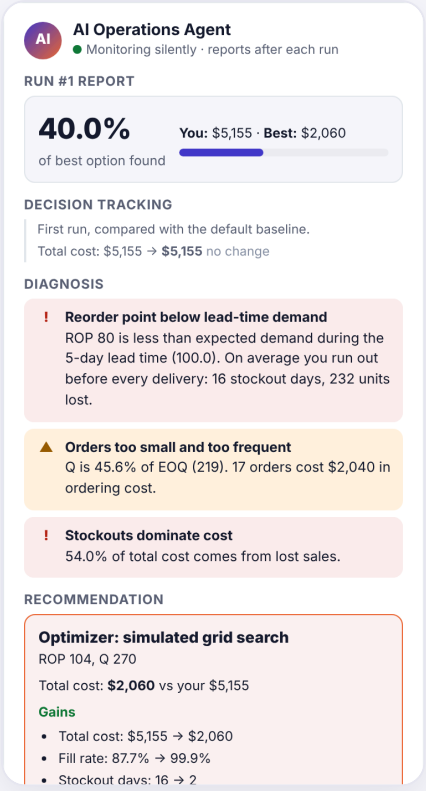
The Optimizer tests hundreds of alternatives on the same scenario: 1,589 inventory policies in about 70 ms.

5 Agent recommends

The best option, with gains and give-ups ("stockout cost – $2,760, holding +$865") and a plain-language explanation.

6 You apply & learn

Apply the recommendation or try your own idea, run again and watch your score climb toward 100% of the best option.



The Operations Coach after one inventory run: diagnosis, best option, gains and give-ups.

What the agent reports, every run

- **Score**: your result as a % of the best option found
- **Decision tracking**: what you changed and what it did
- **Diagnosis**: inefficiencies ranked by severity
- **Recommendation**: the best option with its trade-offs
- **Alternatives tested**: a ranked table you can open
- **Session report**: decisions, mistakes, improvements and lessons, ready to print

Why students trust it: every number comes from the simulation engine. The agent cannot invent a result, and it never changes your inputs on its own.

AGENT SKILLS

12 simulation skills the agents can play and coach

Each skill is a full case: a brief, data exhibits, your decisions, and an optimizer that knows the theory. Figures below are the default case vs the agent's recommendation.

1 Bottleneck & Capacity

Machines per station on a 3-step line.

1/1/1 machines earn $1,340 a shift → 1/2/1 earns $1,920.

2 Inventory Management

Reorder point and order quantity under random demand.

$5,155 and 16 stockout days → $2,060, 99.9% fill rate.

3 Production Scheduling

Dispatching rule or your own job sequence.

FCFS late penalty $22,000 → best of 720 sequences $4,000.

4 Quality Management

Inspection coverage and prevention spend.

Cost of quality $10,220 → $7,497 by preventing, not inspecting.

5 Queue Management

Number of servers vs a service target.

4 servers: $252/h, target missed → 5 servers: $176/h, met.

6 Supply Chain Disruption

Backup supplier share, buffer stock, contingency plan.

$20,976 and 480 units lost → $5,524, 100% fill rate.

7 Process Time Analytics

Batch size, SMED, cell layout, poka-yoke.

Lead time 43 h, $2,601/day → 15 h, $1,068/day.

8 Newsvendor & Contracts

Contract type and one-shot order quantity.

Expected profit $33,304 → revenue sharing: $43,113.

9 Bullwhip Effect

Forecasting, order smoothing, data sharing, lead time.

Factory swings 188× demand, $28,484 → 1×, $8,967.

10 Aggregate Planning

Level, chase or hybrid plan; workforce and overtime.

$4.20M, 84% on time → $3.92M, 99% on time.

11 Network Design

Which of six Indian warehouses to open.

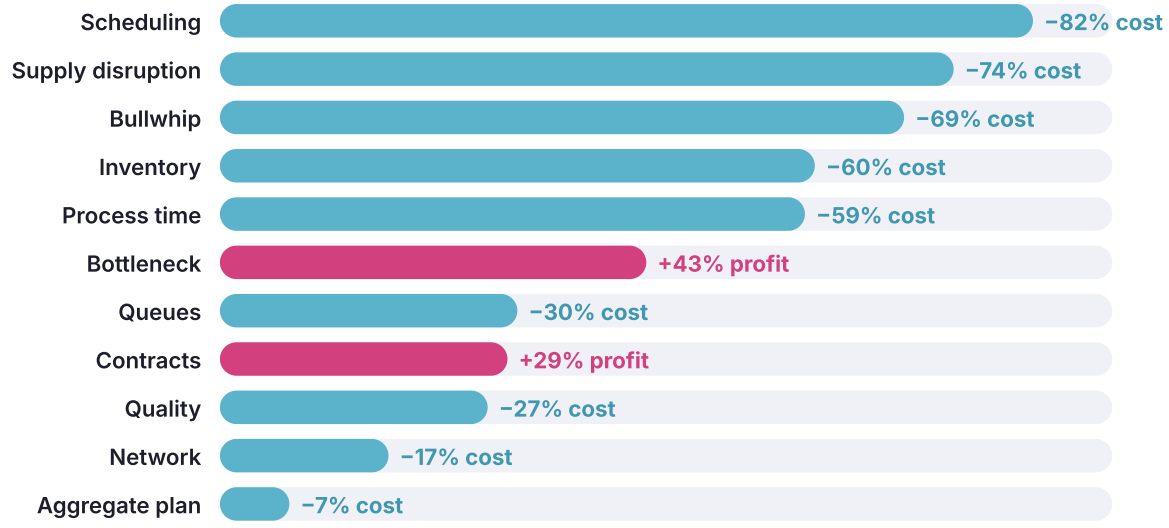
Target missed, $5.29M → 4 DCs meet it at $4.40M.

12 Time Series (ARIMA, SARIMA)

Eight stepwise forecasting decisions.

Every step checked; 10+ models ranked on held-out data.

How much the agents improved each default case



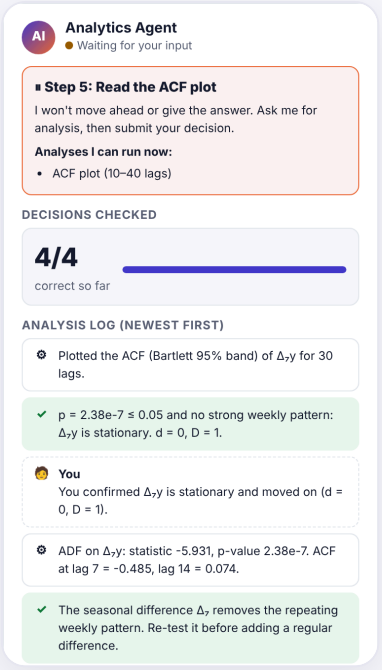
Change in the topic's objective (cost, penalty or profit) from the default decision to the agent's recommendation, on the same scenario. Network design: the default also misses the service target; the agent's network meets it.

SPECIALIST AGENTS IN ACTION

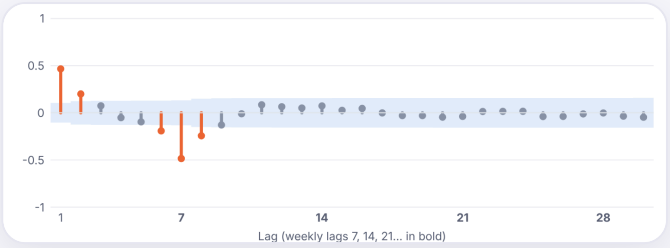
Agents that create, analyse and teach

Analytics Agent

Guides the time-series lab step by step: visualise, test, difference, read the ACF/PACF, fit and choose. It runs only the analysis you ask for, checks each decision, then **waits for you**.



Waiting at Step 5 after four correct decisions on a weekly-seasonal series.



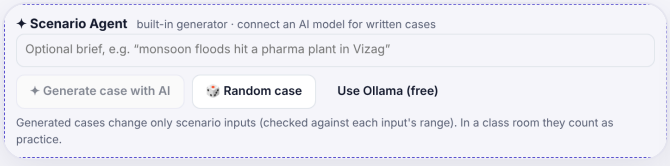
The ACF the agent plotted on request: spikes at lag 1 and lag 7 point to SARIMA.

#	MODEL	FAMILY	PARAMETERS	RMSE	MAPE	
1	**SARIMA(2,0,0)(1,1,1)₇**	ARIMA / SARIMA	6	15.08	2.10%	auto AIC
2	**SARIMA(1,0,0)(0,1,1)₇**	ARIMA / SARIMA	4	15.09	2.17%	your choice, true process
3	**SARIMA(0,0,1)(0,1,1)₇**	ARIMA / SARIMA	4	15.3	2.19%	
4	**Holt-Winters multiplicative**	Holt-Winters multiplicative	13	15.75	2.32%	
5	**Holt-Winters additive**	Holt-Winters additive	13	17.48	2.78%	
6	**Seasonal naïve**	Seasonal naïve (same weekday last week)	0	21.16	3.57%	
7	**Moving average (7 days)**	Moving average	0	151.87	29.17%	
8	**SES**	Simple exponential smoothing (SES)	3	152.48	29.32%	
9	**Naïve**	Naïve (last value)	0	176.64	30.59%	
10	**Holt's linear trend**	Holt's linear trend	5	188.38	32.07%	

The debrief's model tournament: ARIMA, SARIMA, Holt-Winters, SES, naïve and more, ranked on the same test data.

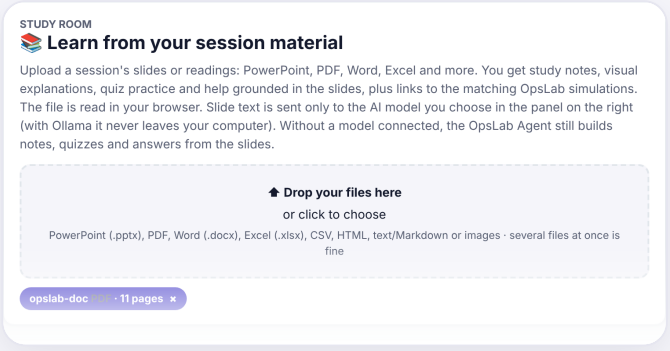
Scenario Agent

Type a brief such as “monsoon floods hit a pharma plant in Vizag”. It writes a new case, or a whole campaign, and checks every value against the allowed limits. It never touches your decisions.



Study Agent

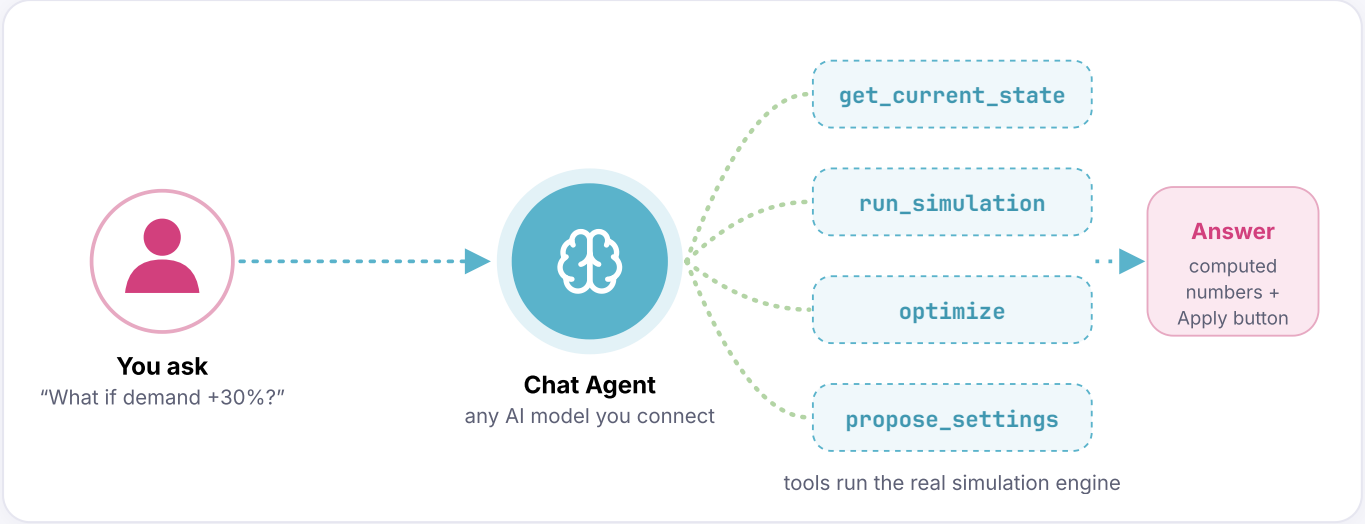
Upload your session slides or readings (PowerPoint, PDF, Word, Excel…). You get study notes, visual explanations, quizzes, slide-grounded help, and links to the matching simulation.



ASK THE AGENT

A tool-using AI agent that tests ideas before answering

Ask in plain words: “Should I add a machine at Assembly?” or “What if demand rises 30%?”. The Chat Agent runs the simulator as its toolbox and answers with computed numbers.



Its four tools

get_current_state
reads your inputs, last run and diagnosis

run_simulation
what-if tests in a sandbox

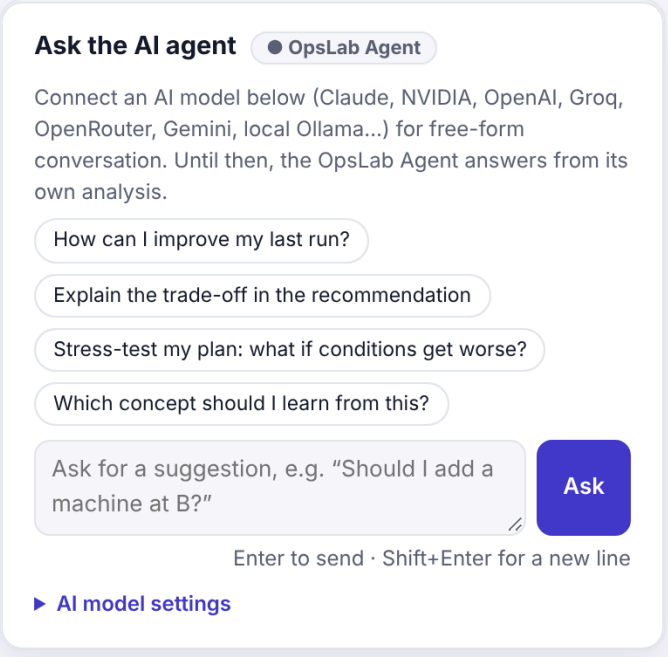
optimize
finds the best decisions, even under a new scenario

propose_settings
shows an “Apply” button: you decide

Connect any AI model

- Anthropic Claude
- NVIDIA NIM
- OpenAI
- Groq
- OpenRouter
- Google Gemini
- Ollama (free, local)
- Any OpenAI-compatible endpoint

Models without tool calling still work: OpsLab hands them the pre-computed simulator data (grounded mode). With no model connected, the OpsLab Agent answers from its own analysis.



The chat box sits under the agent's report on every simulation.

BUILT-IN TRUST · CLASS ROOM

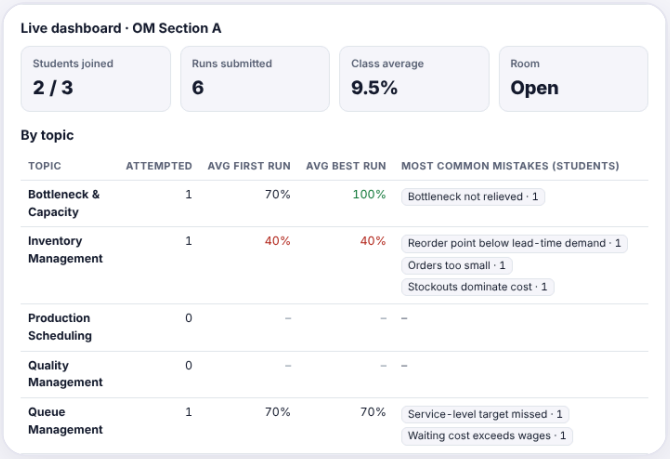
Agents with guardrails, and a whole class in one room

Guardrails every agent follows

- **Computed, not guessed.** KPIs and advice come from the simulation engine.
- **You stay in charge.** Agents suggest; only you apply and run.
- **No interruptions.** Reports appear after a run; advice opens on request.
- **Waits for you.** The Analytics Agent never skips a step or gives the answer.
- **Limits enforced.** Generated cases are checked against every input's range.
- **Private by default.** Files are read in the browser. Text goes only to the AI model you choose; with Ollama it never leaves your laptop.
- **Same answer every time.** Seeded random streams make every run repeatable and fair.

Class Monitor

The professor runs one command on a laptop. Students join with a room code and their roll number. Every run is re-computed by the same engine, so forged scores are ignored. The professor gets a live dashboard: best % per topic, a leaderboard, first-run vs best-run averages, the class's most common mistakes and a CSV for grading.



Professor dashboard (test class).

Campaigns: the agents keep the game going

Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Normal	Growth	Lead time ↑	Festive peak	Calm	Price rise	Shock	Peak
Year 1	Year 1	Year 1	Year 1	Year 2	Year 2	Year 2	Year 2

Play any simulation as 4 or 8 quarters, or 3 or 5 years. Each round brings new conditions: growth, a festive peak, a supplier shutdown revealed only after you commit, a wage deal, a recall. The agents score each round against the best plan.

GET STARTED

Up and running in three steps

1 Open

Open **opslab.html** in Chrome, Edge, Firefox or Safari. No install, no login, works offline.

2 Play

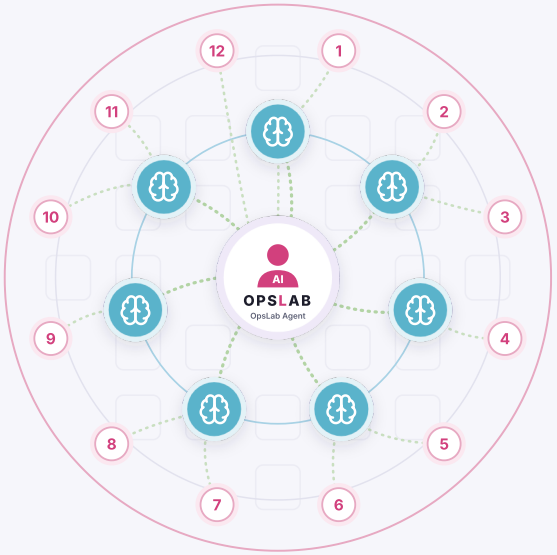
Pick a simulation, read the case data, set your decisions and press **Run**. The agent reports after each run.

3 Level up

Optional: connect an AI model for chat and AI-written cases, upload slides in the Study Room, or start a class room.

A 75-minute class with the agent team

MINUTES	WHAT HAPPENS	AGENTS AT WORK
0–10	Professor opens a class room; students join with roll numbers and read the case brief.	Class Monitor
10–45	Teams play 3–4 simulations: predict, run, read the diagnosis, improve, run again.	Coach, Optimizer
45–60	Stress test: a new case or campaign round changes the world. Is the plan still best?	Scenario Agent, Chat Agent
60–70	Leaderboard and the class's common mistakes on screen; discuss the trade-offs.	Class Monitor
70–75	Print the session report. Homework: time-series lab and a Study Room quiz on today's slides.	Analytics Agent, Study Agent



OpsLab: one page, seven agents, twelve ways to practise operations.

Files: **opslab.html** (the simulator) · **README.md** (quick start) · **docs/OpsLab_Documentation.pdf** (models, formulas and validation)